

Background

Compact high-redshift sources are modelled by subtracting a point-spread function built from nearby field stars. Inferred sizes, luminosities and masses all rest on that subtraction being accurate.

Field-star PSFs assume the target's spectrum matches the stars, which little red dots violate.

What this paper is selling

Using little red dots as a test case, the authors derive the physical and mathematical origin of this spectral mismatch error. They demonstrate a correction applicable to other compact distant sources.

Key results to remember

- A pure point source can masquerade as an extended host in broad filters.
- The spurious F444W host component exceeds the published 11.8% value in every test.
- Sizes, luminosities and masses of little red dots are systematically overestimated.

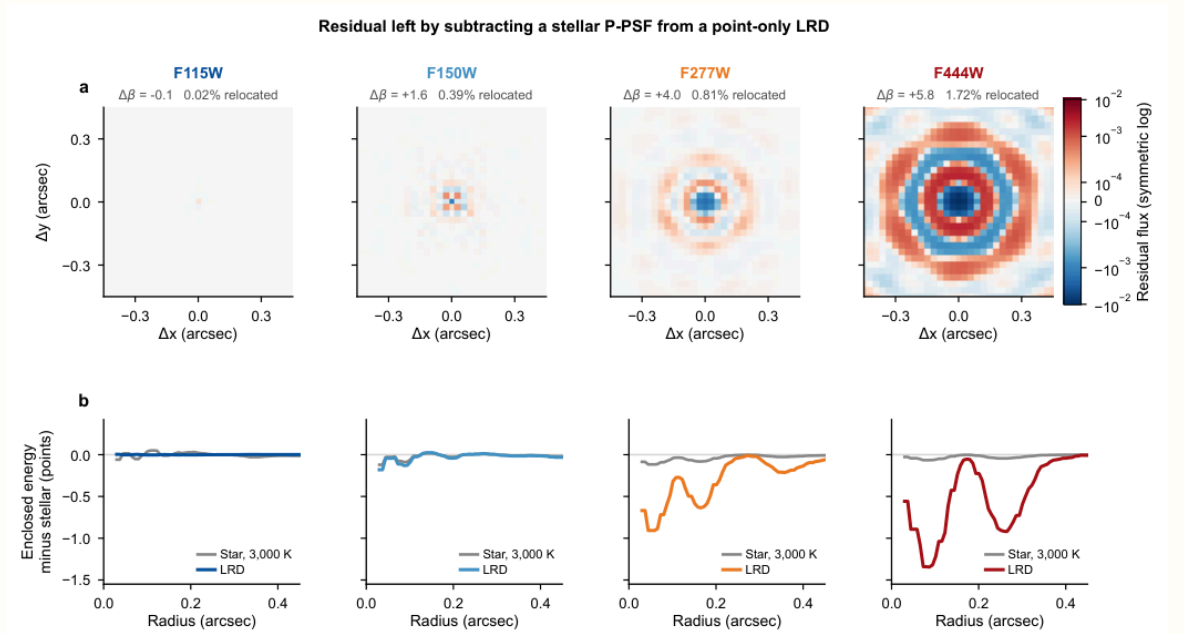


Figure 1 | A point-only LRD produces the wavelength-dependent residual assigned to a Sérsic component.

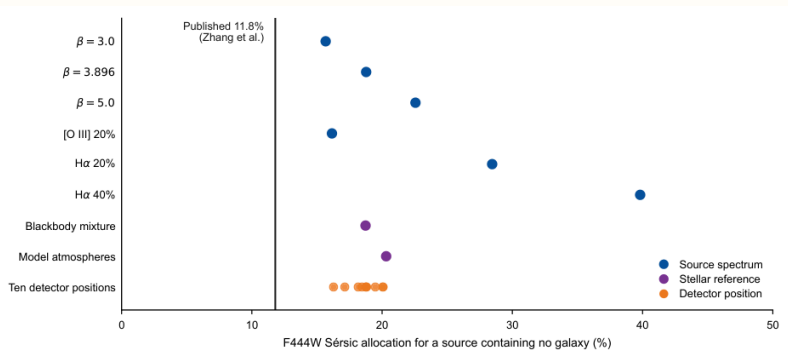


Figure 2 | The F444W allocation for a source containing no galaxy survives every tested change of input.